

ML LAB-2

Name: Harshit Uchasare

Class-Batch: A-A3

Roll no : 35

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv("titanic.csv")
df.head()
```

```
Out[2]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
In [3]: df.tail()
```

```
Out[3]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

```
In [4]: print(df.shape)
```

```
(891, 12)
```

```
In [5]: df.shape
```

```
Out[5]: (891, 12)
```

```
In [6]: df.columns
```

```
Out[6]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',  
              'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],  
              dtype='object')
```

In [7]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [8]: df.describe()

Out[8]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [9]: df.isnull().sum()
```

```
Out[9]: PassengerId      0
Survived      0
Pclass      0
Name      0
Sex      0
Age      177
SibSp      0
Parch      0
Ticket      0
Fare      0
Cabin      687
Embarked      2
dtype: int64
```

```
In [10]: df.head()
```

```
Out[10]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
In [11]: df["Age"].fillna(df["Age"].median(), inplace=True)
```

```
In [12]: df["Embarked"].fillna(df["Embarked"].mode()[0], inplace=True)
```

```
In [13]: df.drop("Cabin", axis=1, inplace=True)
```

```
In [14]: df.head()
```

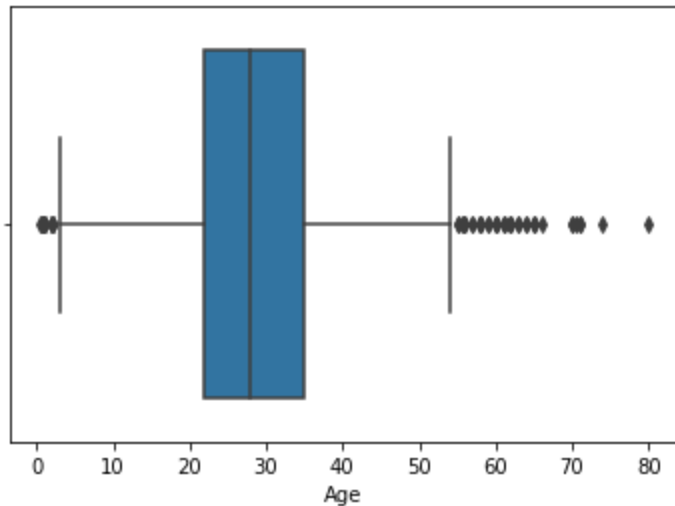
```
Out[14]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S

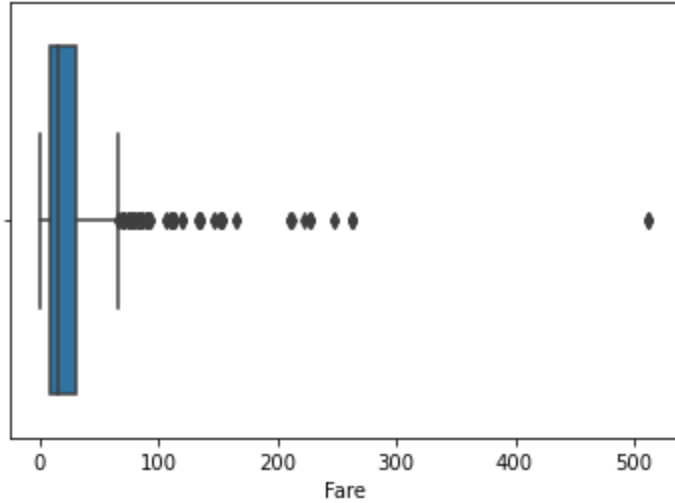
```
In [15]: df.duplicated().sum()
```

```
Out[15]: 0
```

```
In [16]: sns.boxplot(x=df["Age"])  
plt.show()
```

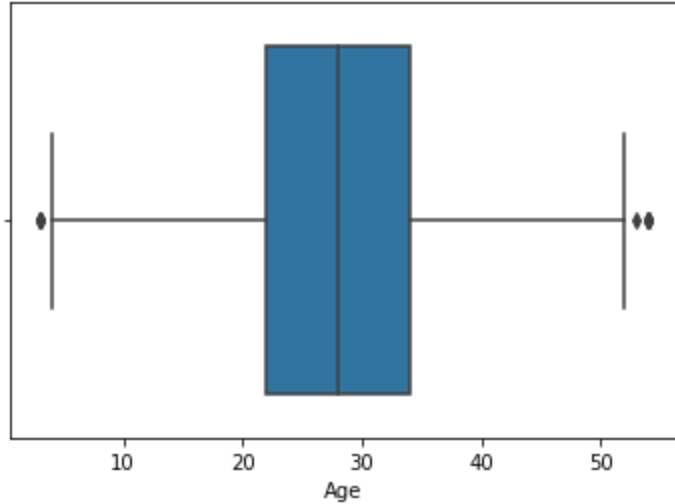


```
In [17]: sns.boxplot(x=df["Fare"])  
plt.show()
```



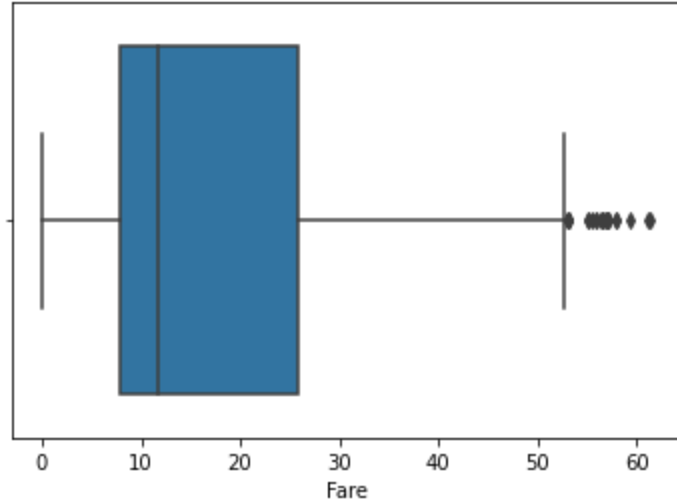
```
In [18]: Q1 = df["Age"].quantile(0.25)  
Q3 = df["Age"].quantile(0.75)  
  
IQR = Q3 - Q1  
  
lower = Q1 - 1.5*IQR  
upper = Q3 + 1.5*IQR  
  
df = df[(df["Age"] >= lower) & (df["Age"] <= upper)]
```

```
In [19]: sns.boxplot(x=df["Age"])  
plt.show()
```



```
In [20]: Q1 = df["Fare"].quantile(0.25)  
Q3 = df["Fare"].quantile(0.75)  
  
IQR = Q3 - Q1  
  
lower = Q1 - 1.5*IQR  
upper = Q3 + 1.5*IQR  
  
df = df[(df["Fare"] >= lower) & (df["Fare"] <= upper)]
```

```
In [21]: sns.boxplot(x=df["Fare"])
plt.show()
```



```
In [22]: df["Sex"] = df["Sex"].replace("male", 0)
df["Sex"] = df["Sex"].replace("female", 1)
```

```
In [23]: df.head()
```

```
Out[23]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	0	22.0	1	0	A/5 21171	7.2500	S
2	3	1	3	Heikkinen, Miss. Laina	1	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	1	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	0	35.0	0	0	373450	8.0500	S
5	6	0	3	Moran, Mr. James	0	28.0	0	0	330877	8.4583	Q

```
In [24]: df["Embarked"] = df["Embarked"].replace("S", 0)
df["Embarked"] = df["Embarked"].replace("C", 1)
df["Embarked"] = df["Embarked"].replace("Q", 2)
```

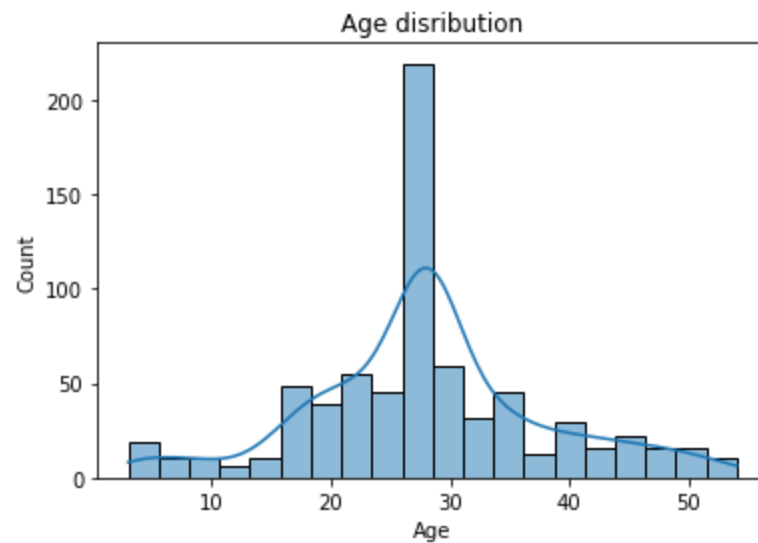
```
In [25]: df.head()
```

```
Out[25]:
```

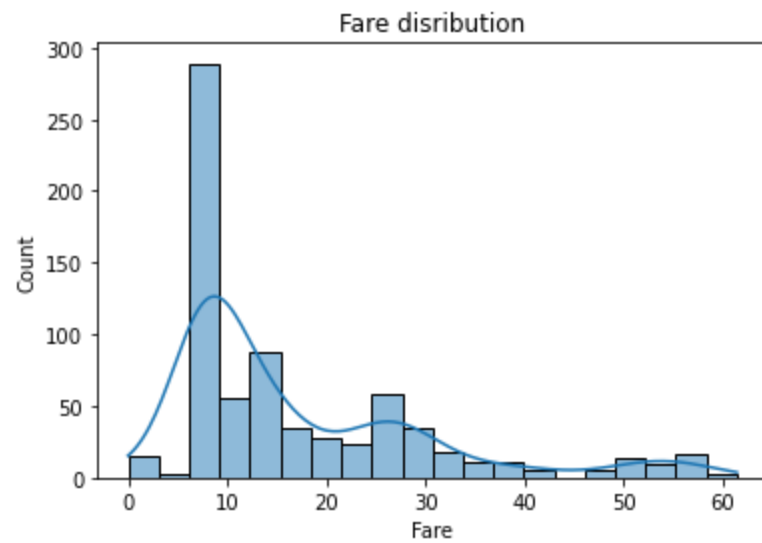
	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	0	22.0	1	0	A/5 21171	7.2500	0
2	3	1	3	Heikkinen, Miss. Laina	1	26.0	0	0	STON/O2. 3101282	7.9250	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	1	35.0	1	0	113803	53.1000	0
4	5	0	3	Allen, Mr. William Henry	0	35.0	0	0	373450	8.0500	0
5	6	0	3	Moran, Mr. James	0	28.0	0	0	330877	8.4583	2

```
In [26]: plt.figure(figsize=(6,4))
sns.histplot(df["Age"], bins = 20, kde=True)

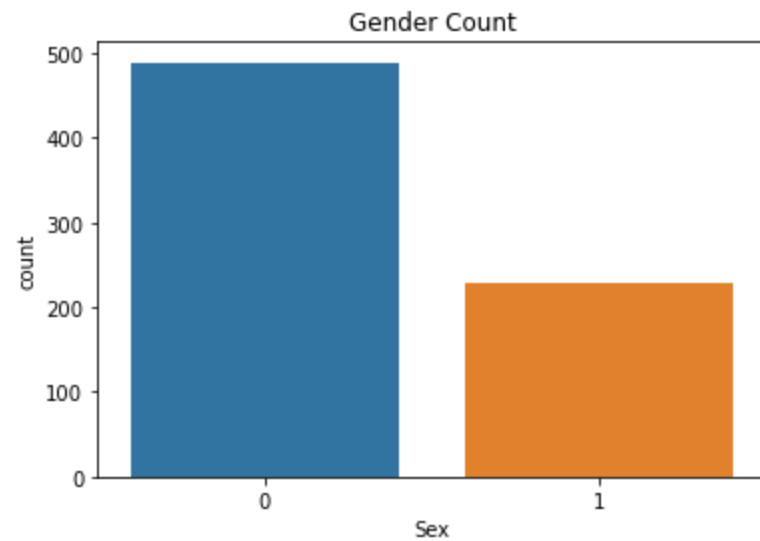
plt.title("Age disribution")
plt.show()
```



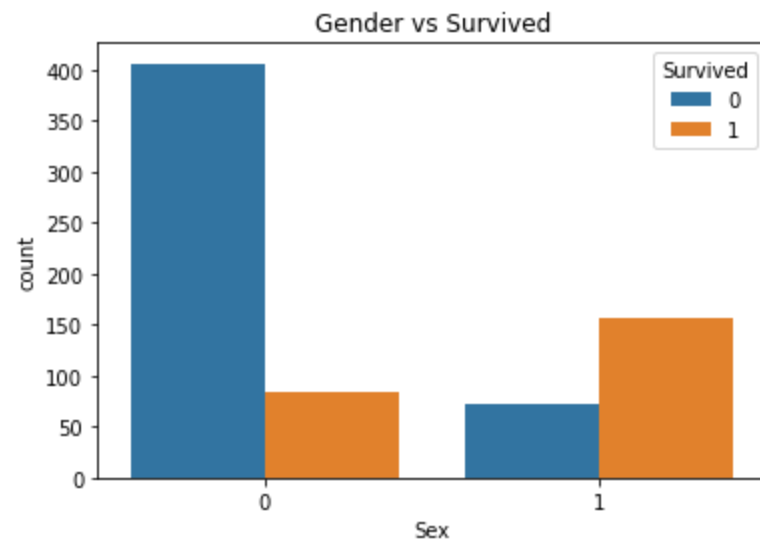
```
In [27]: plt.figure(figsize=(6,4))  
sns.histplot(df["Fare"], bins = 20, kde=True)  
  
plt.title("Fare distribution")  
plt.show()
```



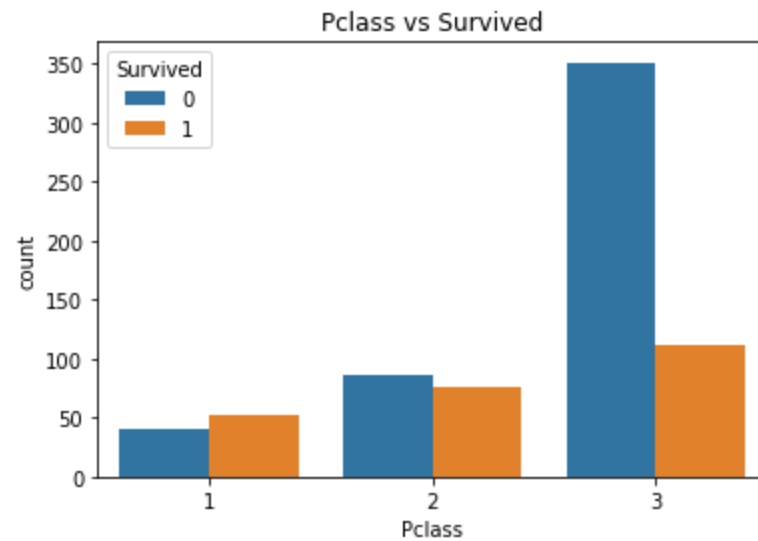
```
In [28]: sns.countplot(x="Sex", data=df)
plt.title("Gender Count")
plt.show()
```



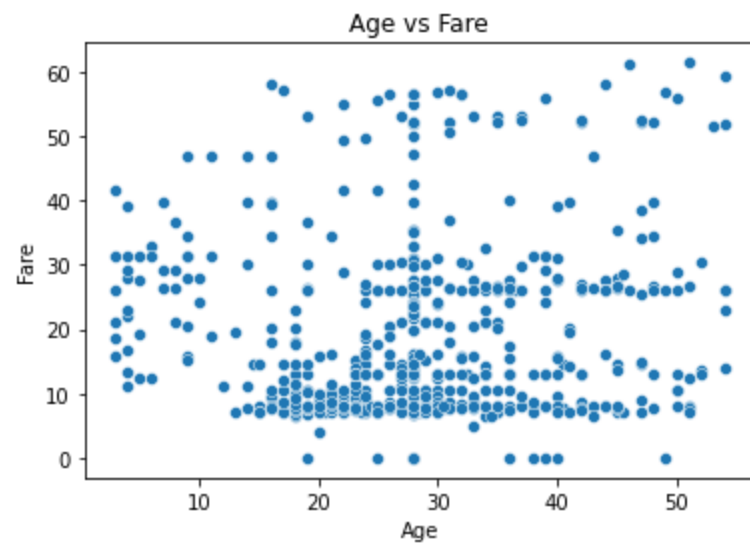
```
In [29]: sns.countplot(x="Sex", hue = "Survived", data=df)
plt.title("Gender vs Survived")
plt.show()
```



```
In [30]: sns.countplot(x="Pclass", hue = "Survived", data=df)
plt.title("Pclass vs Survived")
plt.show()
```



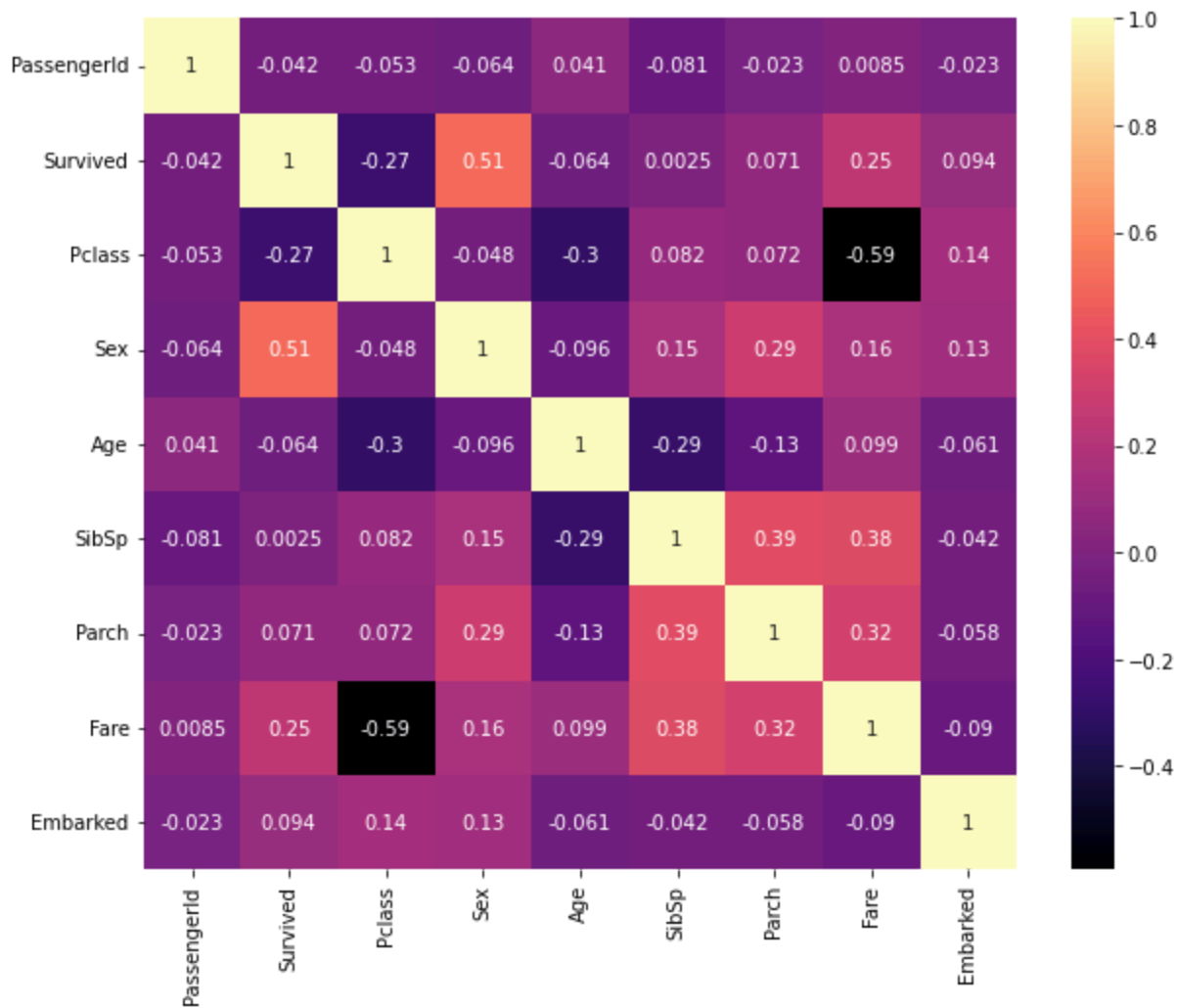
```
In [31]: sns.scatterplot(x="Age", y = "Fare", data=df)
plt.title("Age vs Fare")
plt.show()
```



```
In [32]: plt.figure(figsize=(10,8))
numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(),
            annot=True,
            cmap='magma')

plt.show()
```



```
In [33]: #Feature Scaling
x = df.drop("Survived", axis=1)
y = df["Survived"]
```

```
In [34]: x = x.drop(["PassengerId", "Name", "Ticket"], axis=1)
```

```
In [36]: #Standard Scaling
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

x[["Age", "Fare"]] = scaler.fit_transform(x[["Age", "Fare"]])
```

```
In [37]: x.head(15)
```

```
Out[37]:
```

	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	3	0	-0.607611	1	0	-0.751265	0
2	3	1	-0.207827	0	0	-0.700265	0
3	1	1	0.691688	1	0	2.712961	0
4	3	0	0.691688	0	0	-0.690821	0
5	3	0	-0.007934	0	0	-0.659971	2
6	1	0	2.590664	0	0	2.619461	0
8	3	1	-0.107881	0	2	-0.457860	0
9	2	1	-1.407180	1	0	0.972975	1
10	3	1	-2.406641	1	1	-0.037264	0
12	3	0	-0.807503	0	0	-0.690821	0
13	3	0	1.091473	1	5	1.063959	0
14	3	1	-1.407180	0	0	-0.705614	0
17	2	0	-0.007934	0	0	-0.316820	0
18	3	1	0.291904	1	0	0.060958	0
19	3	1	-0.007934	0	0	-0.753154	1

```
In [38]: #Train Test Split
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.20,random_state=42)
```

```
In [40]: print("Training Data: ", x_train.shape)

print("Testing Data: ", x_test.shape)

Training Data: (574, 7)
Testing Data: (144, 7)
```

```
In [41]: print(x_train.head())
```

	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
69	3	0	-0.207827	2	0	-0.644543	0
248	1	0	0.891580	1	1	2.671722	0
403	3	0	-0.007934	1	0	-0.101487	0
662	1	0	1.891041	0	0	0.634236	0
30	1	0	1.191419	0	0	0.795419	1

```
In [42]: import warnings

warnings.filterwarnings("ignore")
```

```
In [ ]:
```